Math 8 - Surface Area and Volume



— SINE TIMORE AUT FAVORE —

ST. GEORGE'S SCHOOL

Name : _____

Teacher's name: _____

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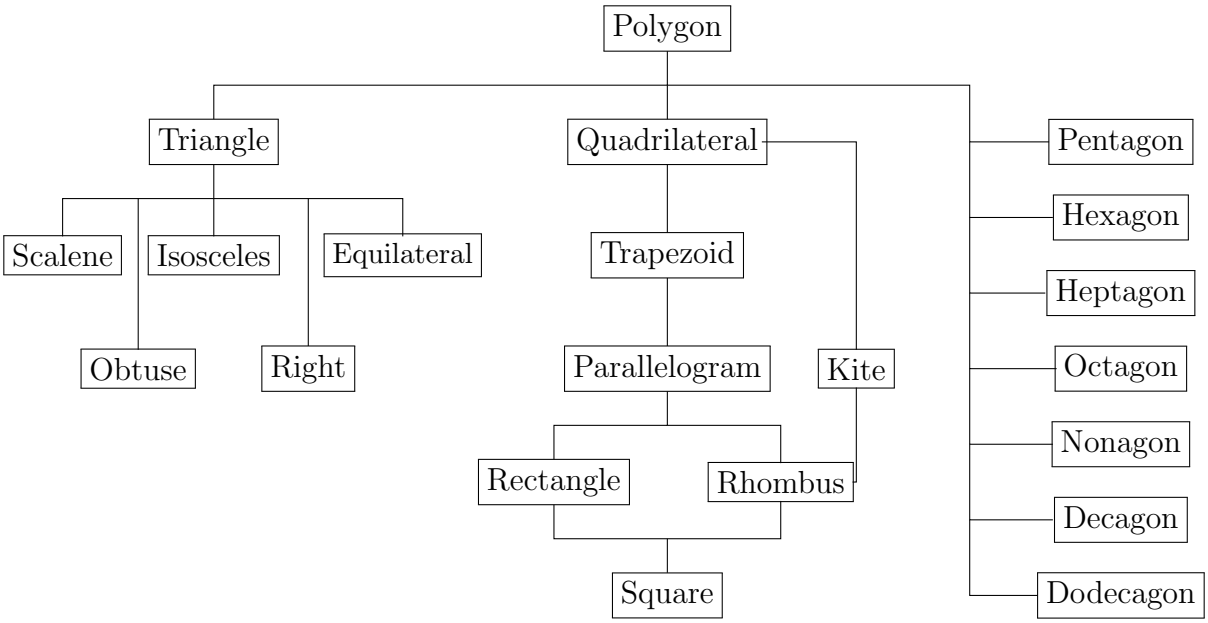
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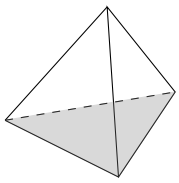
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1 Three Dimensional Figures

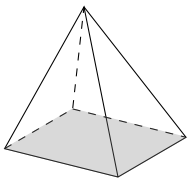
Polygons are closed shapes with any number of straight lines. We have specific names for certain types of polygons



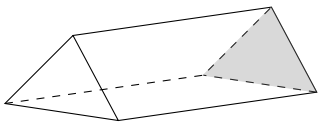
We can compose a three-dimensional object using polygons. This will be called a **polyhedron**.



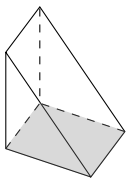
Tetrahedron
(4 faces)



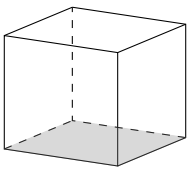
Square Pyramid
(5 faces)



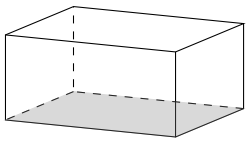
Triangular Prism
(5 faces)



Right Triangular Prism
(5 faces)



Cube
(6 faces)

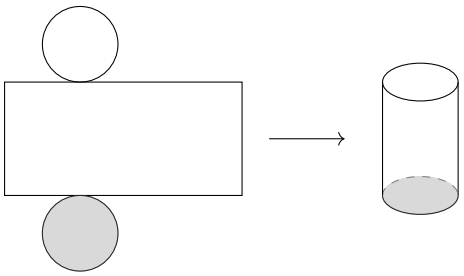


Rectangular Prism
(6 faces)

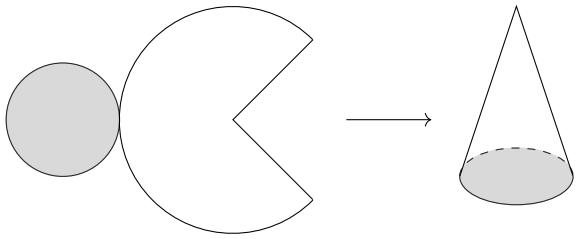
We will be mainly studying right prisms and cylinders, but the ideas we use here are applicable to other shapes as well.

Net Diagrams

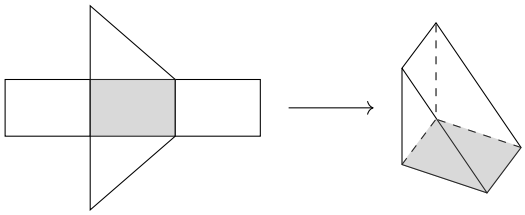
We can use a two-dimensional unfolded representation of a three-dimensional figure.



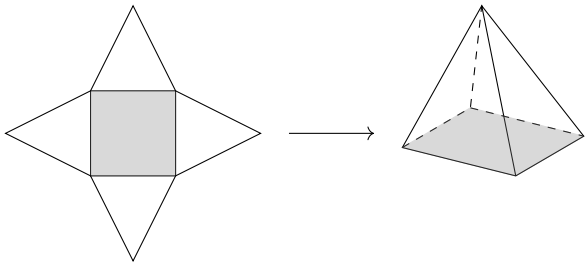
Cylinder



Cone



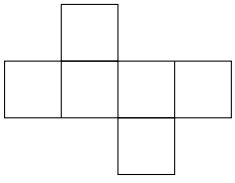
Right Triangular Prism

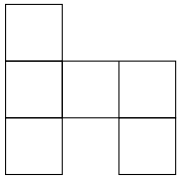


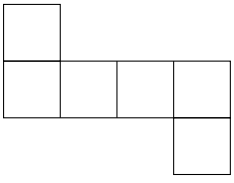
Square Pyramid

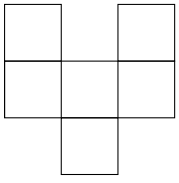
Example 1.1 Draw a net diagram for a cube.

Example 1.2 Which of the following are net diagrams for a cube?

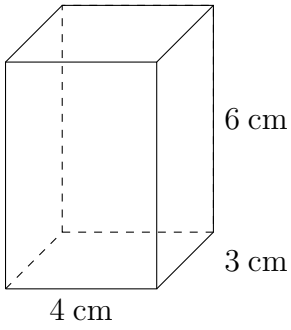
a) 

b) 

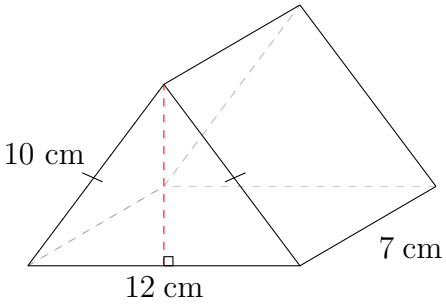
c) 

d) 

Example 1.3 Draw a net diagram of the rectangular prism below and label the length of each side.

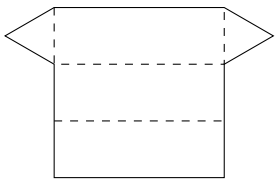


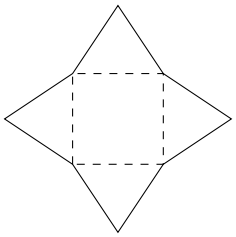
Example 1.4 Draw a net diagram of the triangular prism below and label the length of each side.

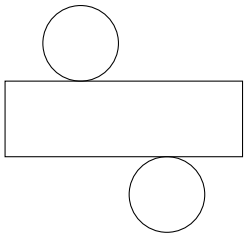


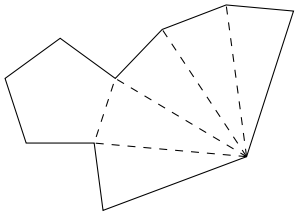
Practice Exercises

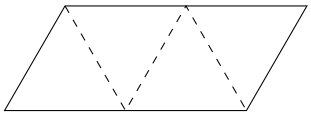
1. For each of the following nets, name the corresponding solid.

a) 

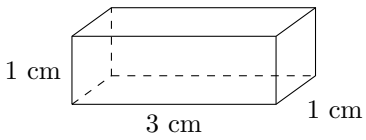
b) 

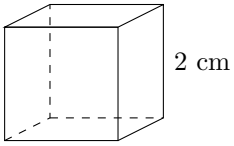
c) 

d) 

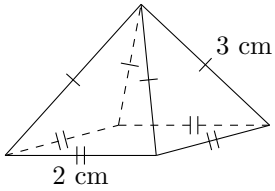
e) 

2. For each of the solids, draw the net diagram with lengths clearly marked.

a) 

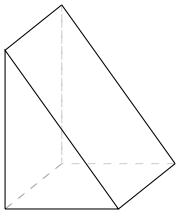
b) 

a cube

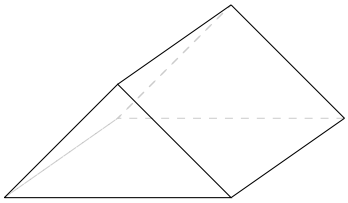
c) 

3. Name the solids below and then draw a net diagram for each. No need to label lengths.

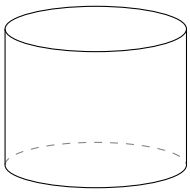
a)



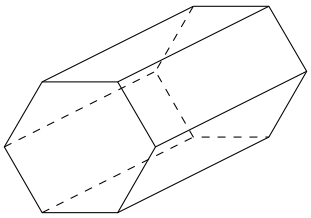
b)



c)



d)



2 Surface Area and Volume of Prisms

The surface area of a three-dimensional object is the sum of the areas of each of the faces of the object. A prism is made from polygons, so formulas for polygons can be used.

Some key terminology:

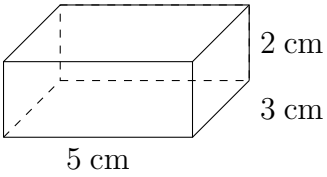
- A **right prism** is one that can be stacked perfectly vertical from the base shape.
- A **regular** polygon is one that has all sides and angles equal.

Surface Area of Rectangular Prisms

Rectangular Prisms have 6 rectangles, where opposite rectangles are equal.

$$SA = 2lw + 2lh + 2wh = 2(lw + lh + wh)$$

Example 2.1 Find the surface area of the below rectangular prism.



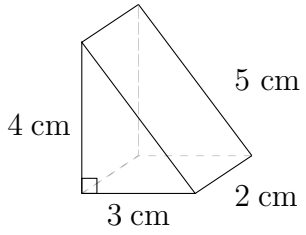
Surface Area of Triangular Prisms

Triangular prisms have a triangle base (can be a stack of triangles).

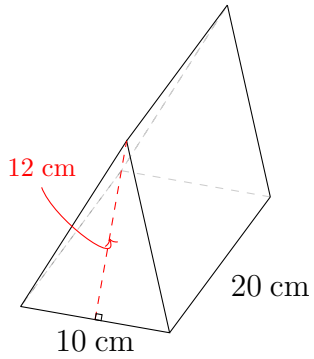
Useful triangle formulas

$$Area = \frac{1}{2} \times base \times height \qquad a^2 + b^2 = c^2$$

Example 2.2 Find the surface area of the below right-triangular prism.



Example 2.3 Find the surface area of the below prism.



Volume of Prisms

Volume is a three-dimensional measure, so the units will always be in cubic units.

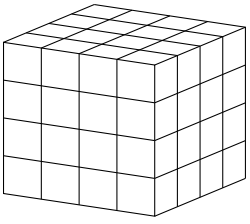
Prisms will always have a base shape that is the same on the bottom as it is on the top.

The other surfaces wrapping around the base are often referred to as **lateral surfaces**.

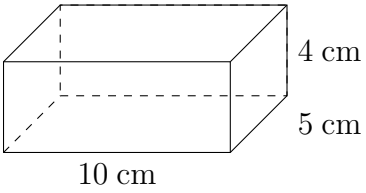
For right prisms, the volume is simply

$$V = (base\,area) \times (height)$$

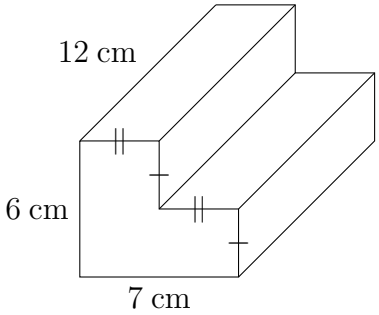
Example 2.4 Each of the smaller cubes have a volume of 1 cubic centimeter. Find the volume of all the cubes.



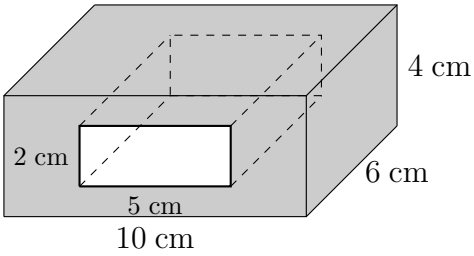
Example 2.5 Find the volume of the following rectangular prism.



Example 2.6 Find the volume of the prism.



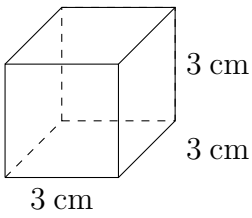
Example 2.7 Find the volume of the prism.



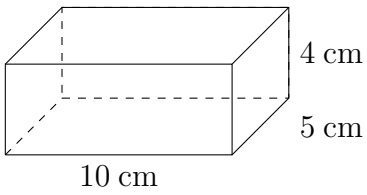
Practice Exercises

1. Find the surface area of the below prisms.

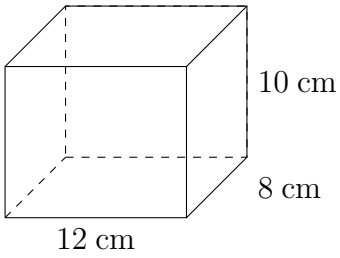
a)



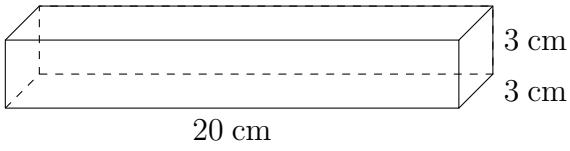
b)



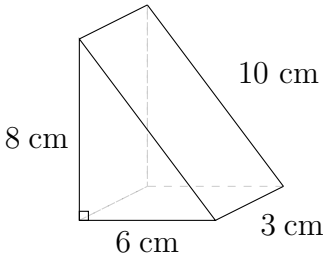
c)



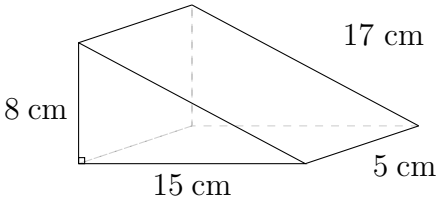
d)



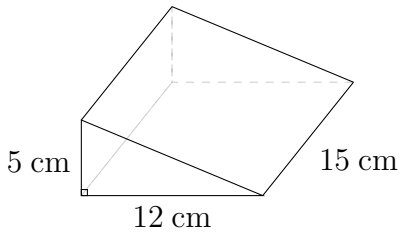
e)



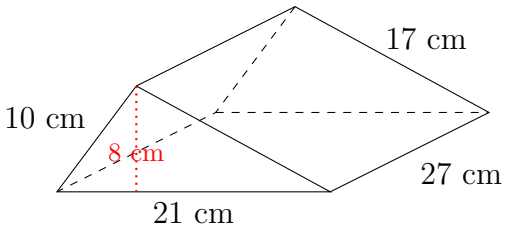
f)



g)

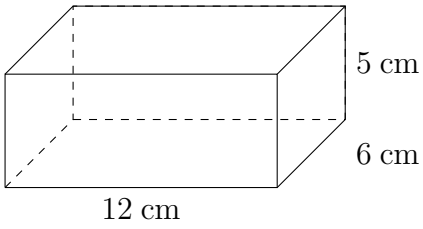


h)

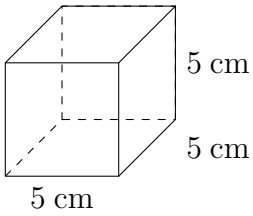


2. Find the volume of the prisms.

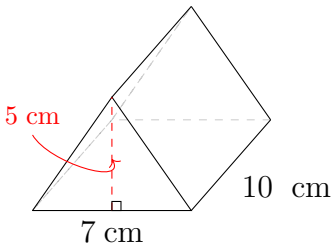
a)



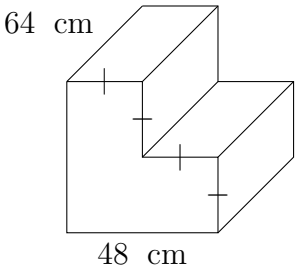
b)



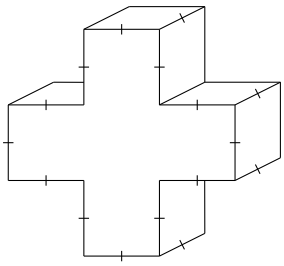
c)



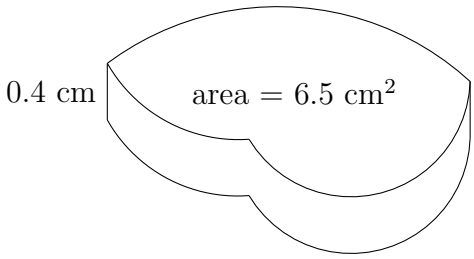
d)



e)



f)

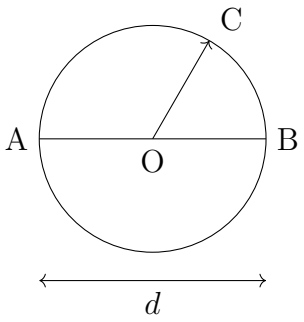


3. Complete the table of measurements for a rectangular prism.

	l	w	h	SA	V
a)	8	5	3		
b)	25	12	10		
c)	8	4			192
d)		3	9		189

3 Surface Area and Volume of Cylinders

A cylinder is shaped like a canned food item that you can find in stores. It would not be considered a polyhedron because it has a circle as the base.
A circle has several properties we commonly refer to.



- It has a center which is shown by the point O above
- It has a diameter, which is the length of the line that crosses the center (d above)
- It has a radius, which is the length from the center to any point on the circle (r above)

The **radius** is always half the **diameter**, or the diameter is always double the radius.

Circumference of a Circle
The circumference of the circle is the distance around the circle (like perimeter).

$$C = \pi d \quad or \quad C = 2\pi r$$

Example 3.1 Find the circumference of a circle with diameter 6 cm.

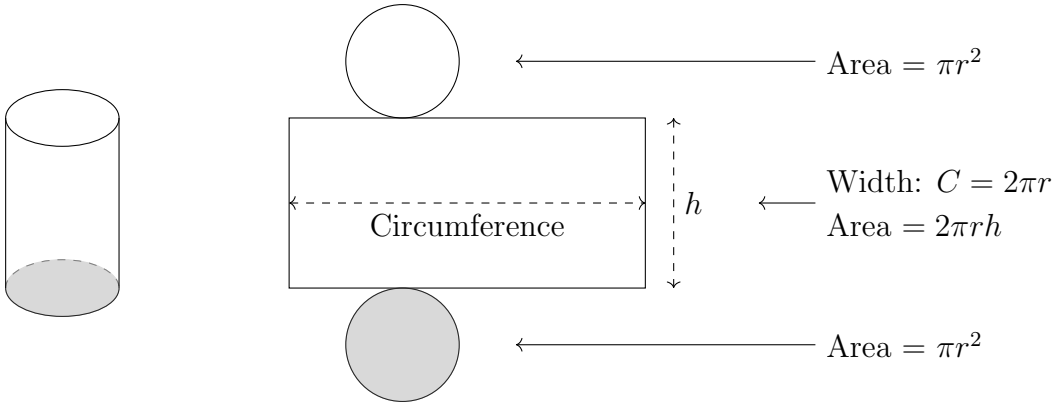
Area of a Circle
The area of the circle is the 2-dimensional space it takes up.

$$A = \pi r^2$$

Surface Area of a Cylinder
The surface area of a cylinder is

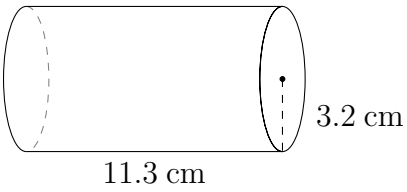
$$SA = 2\pi r^2 + 2\pi rh$$

We can look at the net diagram of a cylinder to see how the surface area formula is derived.



Example 3.2 Find the surface area of a cylinder with a diameter of 12 cm and a height of 5 cm.

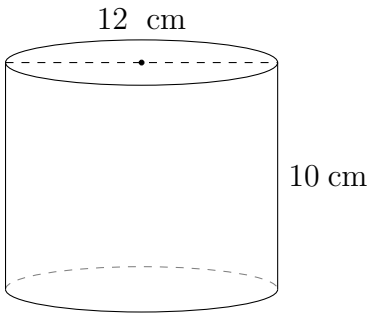
Example 3.3 Find the surface area of the following.



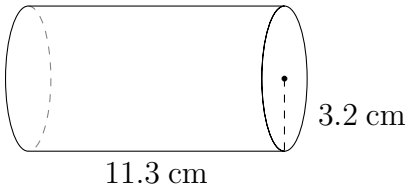
Volume of Cylinders
The volume of a cylinder is similar to the volume of a prism, as both the base and top surfaces are the same. It is essentially a vertical stack of circles.
Hence,

$$V = (base\,area) \times height \quad \rightarrow \quad V = \pi r^2 h$$

Example 3.4 Find the volume of the following.



Example 3.5 Find the volume of the following.



Practice Exercises

1. Find the surface area and volume of a cylinder with the following measurements. Leave answers in terms of π. All units are in cm.

   a) $r = 4$
   $h = 6$

   b) $r = 5$
   $h = 9$

   c) $r = 7$
   $h = 6.5$

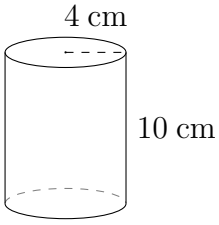
   d) $d = 10$
   $h = 8$

   e) $d = 19$
   $h = 6$

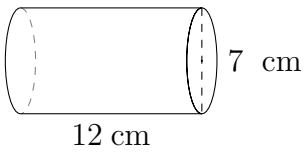
   f) $d = 10.5$
   $h = 9.4$

2. Find the surface area and volume of the following solids. Round answers to the nearest hundredths.

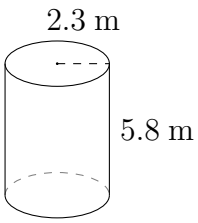
   a)



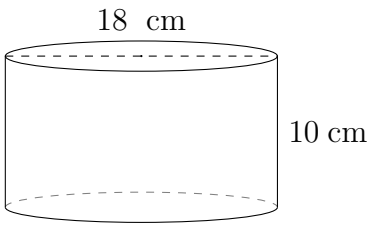
   b)



c)



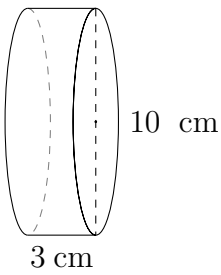
d)



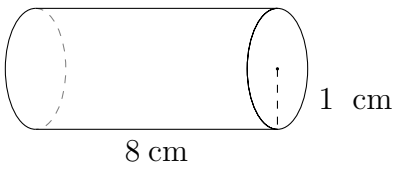
3. In your own words, define the lateral area of a cylinder. What would be the formula to calculate the lateral area?

4. Find the lateral area of the following cylinders.

a)



b)



4 Surface Area of Composite Solids

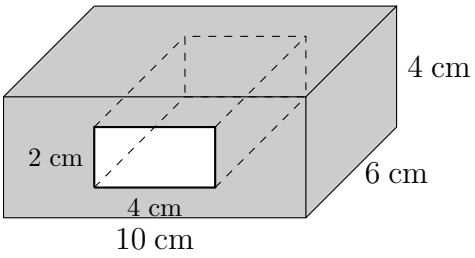
Finding volume of composite solids is straight forward, as we can simply add the different solids or minus the different solids' volumes. However, surface area of composite solids are more complicated.

For some shapes that are composite of multiple shapes, there are some good strategies that can work on most common cases.

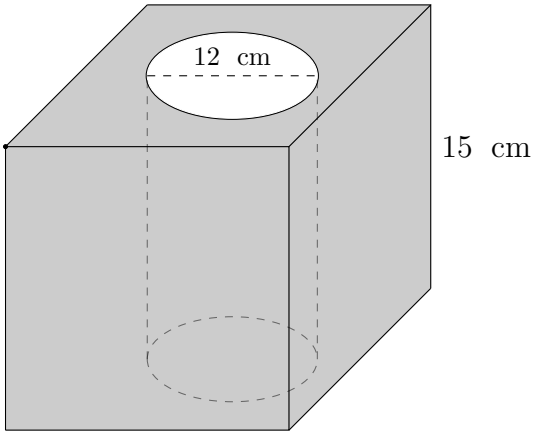
Solids with Missing Centres

These solids have volume that is the original minus the inside, however, with surface area, we lose the two ends but gain surface inside the solid.

Example 4.1 Find the surface area of the prism with a hole the shape of a prism cut through.



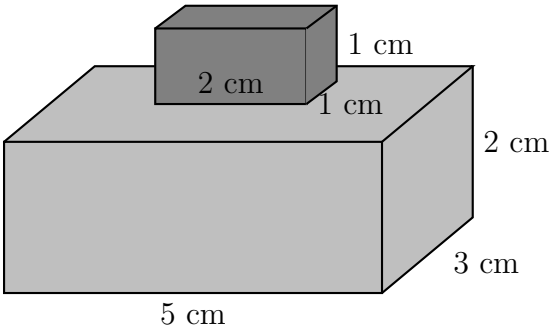
Example 4.2 Find the surface area of a 15 cm cube with a cylinder of a diameter 12 cm removed from its center.



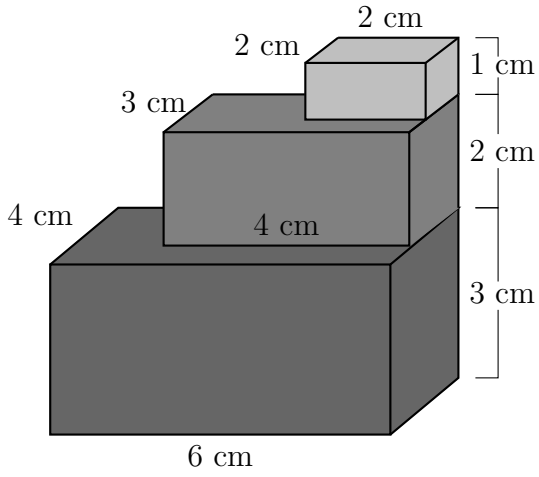
Front/Back - Side/Side - Top/Bottom Strategy

For most flat shapes, you only need to calculate three different surfaces and then double each one. The view perspective of the front/back, side/side, top/bottom looks the same from a two-dimensional standpoint. This **does not work** when there are slanted or curved surfaces, or any inside surfaces.

Example 4.3 Find the surface area of a rectangular prism stacked on top of another rectangular prism.



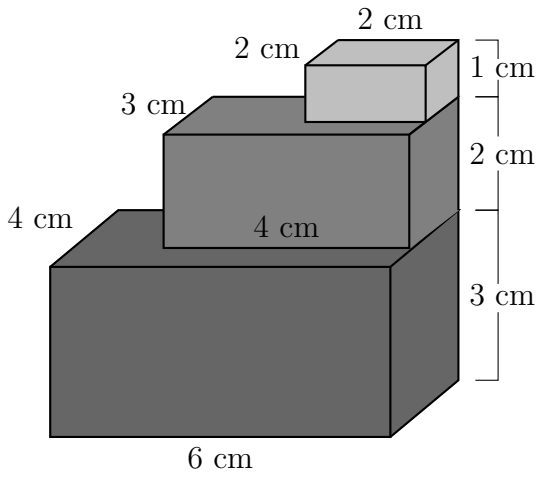
Example 4.4 Find the surface area of three rectangular prisms stacked on top of one another.



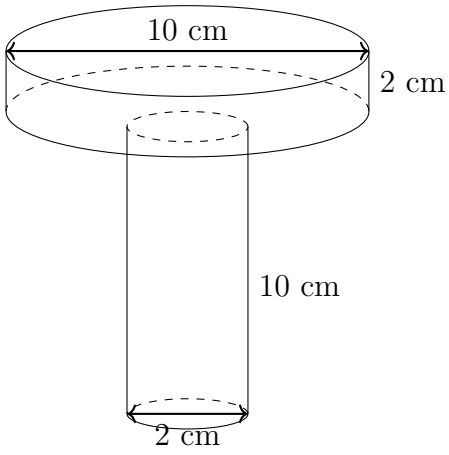
Minus Twice the Overlap Strategy

There is another valid strategy that might work better if you like formulas. We can calculate each shape's surface area, which is typically over-counting. Then we can minus twice any overlapped surface, to account for the initial over-count.

Example 4.5 Find the surface area of the following.



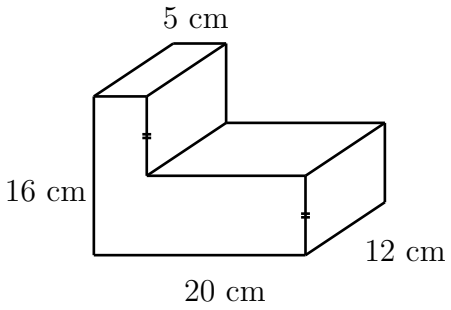
Example 4.6 Find the surface area of the following.



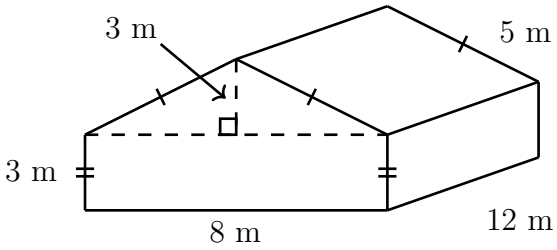
Practice Exercises

1. Find the surface area of the composite solids.

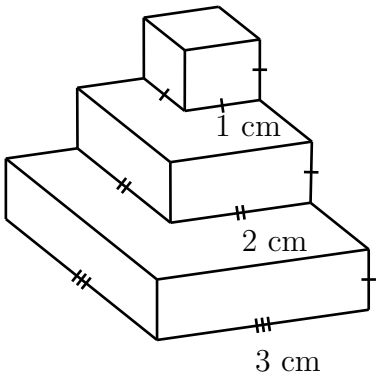
a)



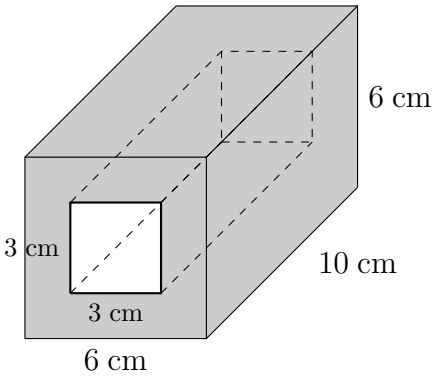
b)



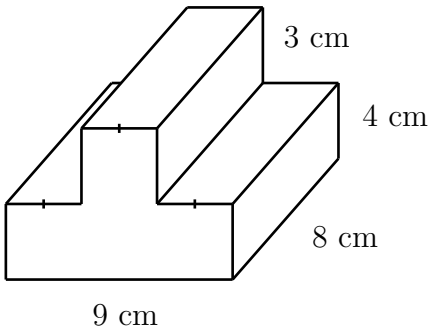
c)



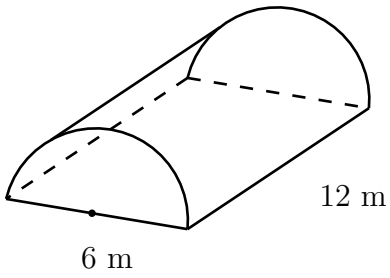
d)



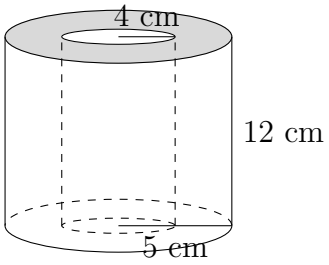
e)



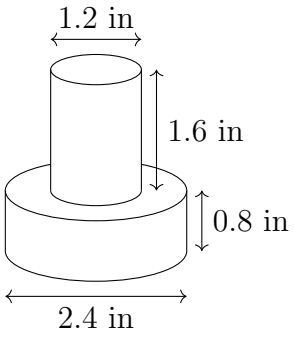
f)



g)



h)



5 Applications and Miscellaneous Shapes

We can look at these formulas from an algebra perspective by trying to solve for an unknown.

Example 5.1 If a right prism has a base area of 25 cm^2 and a volume of 200 cm^3, what is the height of the prism?

Example 5.2 What is the length of a rectangular prism with volume 420 cm^3, with width 6 cm and height 10 cm?

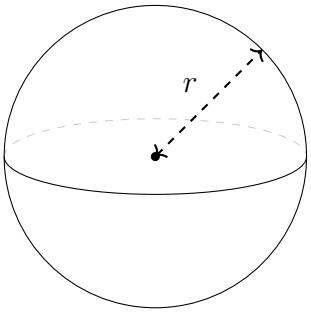
Example 5.3 What is the width of a rectangular prism if the surface area is 310 cm^2 and the length and height are 10 cm and 5 cm respectively.

Example 5.4 The volume of a cylinder is 8π cm^3. If $r = 2$ cm, find h.

Example 5.5 The volume of a cylinder is 64π in^3. If $r = h$ cm, find r.

There are other solids that you should familiarize yourself with, such as spheres, cones, and pyramids. We'll look at the sphere and cone, and the formulas that are associated with them.

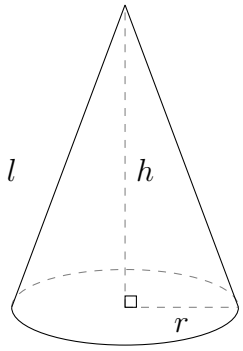
Sphere



$$SA = 4\pi r^2$$

$$V = \frac{4}{3}\pi r^3$$

Cone

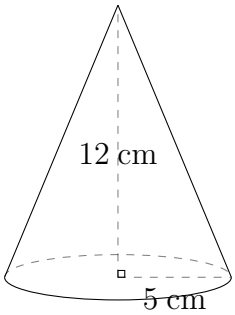


$$SA = \pi rl + \pi r^2$$

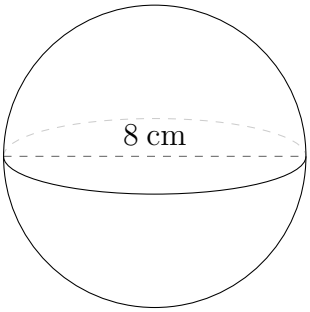
$$V = \frac{1}{3}\pi r^2 h$$

Example 5.6 Find the surface area and volume of the following.

a)



b)



Practice Exercises

1. What is the height of a rectangular prism with length 5 cm, width 7 cm, and volume 140 cm^3?

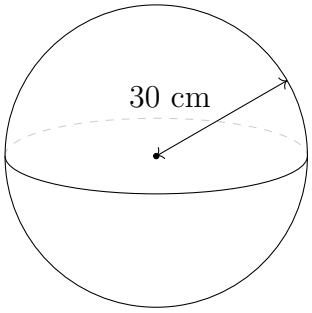
2. The volume of a cylinder is 45π cm^3. If $r = 3$ cm, find h.

3. A rectangular prism has a square base and a height of 6 in. If the volume of the prism is 150 in^3, what is the length of the square base?

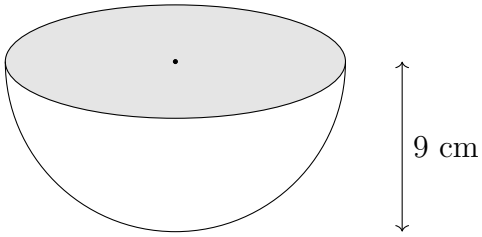
4. A rectangular prism with surface area 168 cm^3 has a width of 4 cm, and a height of 8 cm. What is the measurement of the length?

5. Find the surface area and the volume of the following solids.

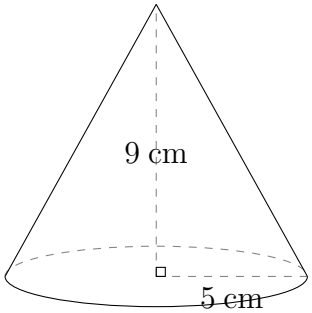
a)



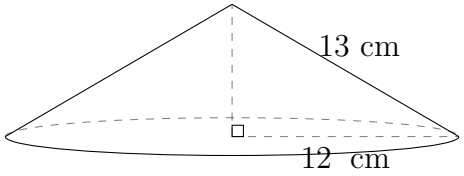
b)



c)



d)



6 Reasoning and Analyzing Problems

Each statement is either **always true, sometimes true,** or **never true**. When you decide on your answer, you must provide a **justification**.

If you choose **sometimes true**, you must show one scenario when it is true, and another, when it is false.

For **always true** and **never true**, you must provide a brief explanation why. Sometimes this can simply be a definition, or sometimes it may require some more mathematical reasoning.

Example 6.1 For each of the following statements, identify if the statement is always true, sometimes true, or never true. Provide a brief justification.

a) A regular polygon has all interior angles equal.

b) A pyramid is a prism.

c) If the figure is a rectangular prism, then it is a cube.

d) If the figure is a cube, then it is a rectangular prism.

e) The lateral surface of a cylinder is a rectangle.

f) The volume of a prism is the product of the base area and the height.

g) The volume and the surface area of an object are not equal.

Practice Exercises

1. For each of the following statements, identify if the statement is always true, sometimes true, or never true. Provide a brief justification.

   (a) A rectangle is a parallelogram.

   (b) A parallelogram is a rectangle.

   (c) A right-triangular prism has three distinct rectangles.

   (d) The surface area of a cylinder with an open top is $SA = \pi r^2 + 2\pi rh$ where r and h represent the radius of the circle and height of the prism.

   (e) The area of a circle in terms of the diameter is $A = \dfrac{\pi d^2}{2}$.

   (f) The area of a circle in terms of the diameter is $A = \pi \left(\dfrac{d}{2}\right)^2$.

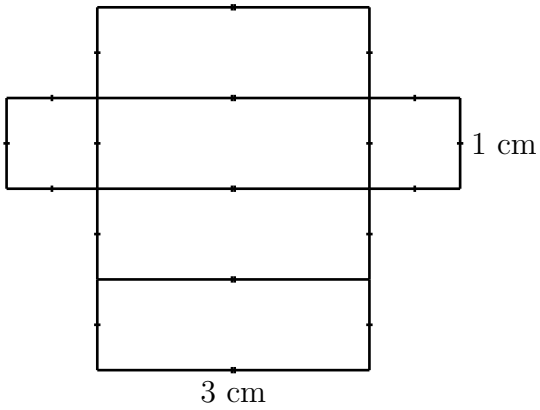
Answer Key

Section 1 - Exercises

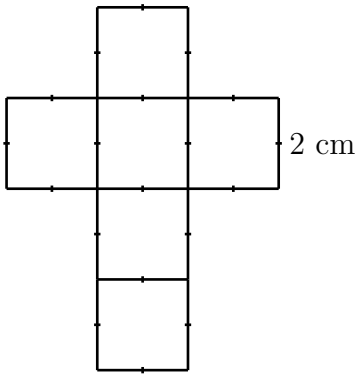
1. a) Triangular Prism b) Square Pyramid c) Cylinder
   d) Pentagonal Pyramid e) Tetrahedron

2. Net diagrams may vary slightly.

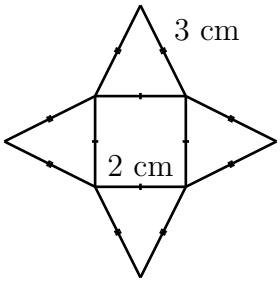
a)



b)

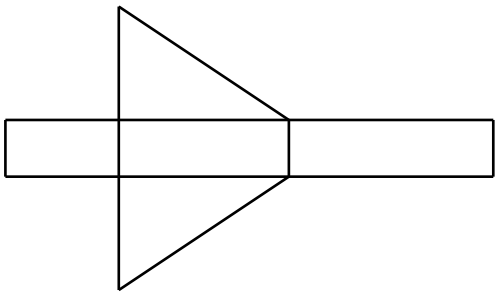


c)

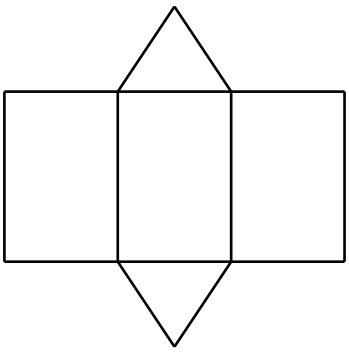


3. Net diagrams may vary slightly.

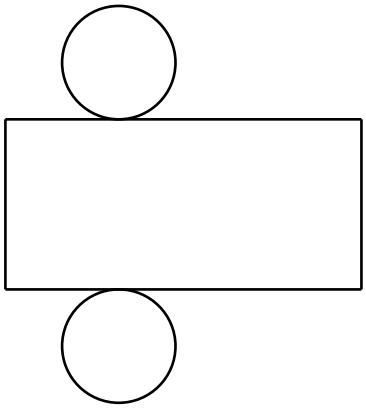
a) Right Triangular Prism



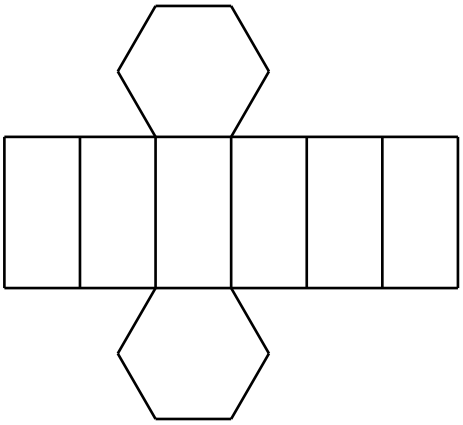
b) Triangular Prism



c) Cylinder



d) Hexagonal Prism



Section 2 - Exercises

1. a) 54 cm^2 b) 220 cm^2 c) 592 cm^2 d) 258 cm^2
   e) 120 cm^2 f) 320 cm^2 g) 510 cm^2 h) 1464 cm^2

2. a) 360 cm^3 b) 125 cm^3 c) 175 cm^3
   d) 110592 cm^3 e) 320 cm^3 f) 2.6 cm^3

3. a) 158, 120 b) 1340, 3000 c) 6, 208 d) 7, 222

Section 3 - Exercises

1. a) $SA = 80\pi\, cm^2$
   $V = 96\pi\, cm^3$
   b) $SA = 140\pi\, cm^2$
   $V = 225\pi\, cm^3$
   c) $SA = 187\pi\, cm^2$
   $V = 318.5\pi\, cm^3$
   d) $SA = 130\pi\, cm^2$
   $V = 200\pi\, cm^3$
   e) $SA = 294.5\pi\, cm^2$
   $V = 541.5\pi\, cm^3$
   f) $SA = 153.825\pi\, cm^2$
   $V = 259.0875\pi\, cm^3$

2. a) $SA \approx 351.86\, cm^2$
   $V \approx 502.65\, cm^3$
   b) $SA \approx 340.86\, cm^2$
   $V \approx 461.81\, cm^3$
   c) $SA \approx 117.06\, cm^2$
   $V \approx 96.39\, cm^3$
   d) $SA \approx 1074.42\, cm^2$
   $V \approx 2544.69\, cm^3$

3. The lateral area of a cylinder is a rectangle that wraps around the side.
   Formula = Circumference $\times$ Height $= (2\pi r)h$

4. a) $30\pi\, cm^2 \approx 94.25\, cm^2$ b) $16\pi\, cm^2 \approx 50.27\, cm^2$

Section 4 - Exercises

1. a) $1264\,cm^2$ b) $360\,m^2$ c) $42\,cm^2$ d) $414\,cm^2$
   e) $346\,cm^2$ f) $(72+45\pi)\,cm^2 \approx 213.37\,cm^2$ g) $234\pi\,cm^2 \approx 735.13\,cm^2$ h) $6.72\pi\,in^2 \approx 21.11\,in^2$

Section 5 - Exercises

1. $h = 4\,cm$
2. $h = 5\,cm$
3. Base is $5\,in.$
4. $l = \frac{13}{3}\,or\,4\frac{1}{3}\,cm$
5. a) $SA = 3600\pi\,cm^2 \approx 11309.73\,cm^2$
   $V = 36000\pi\,cm^3 \approx 113097.34\,cm^3$
   b) $SA = 243\pi\,cm^2 \approx 763.41\,cm^2$
   $V = 486\pi\,cm^3 \approx 1526.81\,cm^3$
   c) $SA \approx 240.26\,cm^2$
   $V = 75\pi\,cm^3 \approx 235.62\,cm^3$
   d) $SA = 300\pi\,cm^2 \approx 942.48\,cm^2$
   $V = 240\pi\,cm^3 \approx 753.98\,cm^3$

Section 6 - Exercises

1. a) Always true. A parallelogram must have opposite sides parallel, which a rectangle does.
   b) Sometimes true. A parallelogram where adjacent sides do not meet at 90° will not be a rectangle. But if they do, then it is a rectangle.
   c) Sometimes true. If the right triangle is scalene (three different sides), then it will have three distinct rectangles. However, if the right triangle was isosceles (two equal sides), then two of the rectangles will be identical.
   d) Always true. An open top means one area of a circle less, so the formula stands.
   e) Never true. dividing by 2 after taking the exponent is the incorrect and produces a different answer.
   f) Always true. Dividing the diameter by 2 first, and then raising it to the power of 2 is the correct order.